Beyond few harmonics: the lower side of the logarithmic law of the resolution gap for windows with a weak ripple

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Paper F [1] proved the logarithmic law of the resolution gap with the sharp constant for periodic windows with at most twenty harmonics or with decay exponent $K \geq 0.003$ (which includes every window with $x \geq 0.003$): with $\ell = \ln(1/\sigma)$, the smallest gap obeys explicit two-sided bounds on $g_{\min}\sqrt{\ell}$ for $10^{-100000} \leq \sigma \leq 10^{-24}$, and its limit lies between $1 - \bar{\varepsilon}$ and 1, where $\bar{\varepsilon}$ bounds the remainder on the window class. Here we extend the lower side to windows with any number of harmonics whose ripple is weak in the sup norm or poor in low-frequency energy. Three tools do the work. A sharper treatment of one remainder of the curvature step turns an absolute error of size $e^{-\tau/2}/k'$ into a relative one. A split sandwich lets a low-frequency block enter through its energy rather than through its sup. Cell-based computer-assisted proofs, with an explicit bound on floating-point rounding in every acceptance test, certify the full lower bound, remainders included, and a mean-square bound of the curvature cross term closes the edge of very wide focus. The main theorem gives $g\sqrt{\ell} \geq (1 - \bar{\varepsilon})(\lambda/\Upsilon)(1 + 12.1/\ell + 2.5 \times 10^{-4})$ for every state with $\tau \geq 20$, for $\ell \in [55.26, 960.8]$, that is, for σ from 10^{-24} down to about 10^{-417}. In this range the coefficient one holds, up to the factor $1 - \bar{\varepsilon}$, for every window with $\eta \leq 1/\ell$, for a family of intermediate windows, and for seventeen thin-block windows certified one by one. A conjecture of our own (that $\eta e^{\mu/2}$ stays bounded) is false; we give explicit counterexamples, and they are covered by the theorem. We close with the road to an unconditional theorem, from the nearest restriction to the farthest. For the states with a narrow focus ($\tau < 20$) everything reduces to one local inequality, which holds within a factor 2.13 on the windows we tried while the bound tolerates a factor 10. Further out lie the full bound beyond $\ell = 960.8$ and the limit $\sigma \to 0$, windows outside the covered families, the class constant $\bar{\varepsilon}$, and the upper side. We list exactly what is proved, what is computer-assisted, and what is only numerical.

I. INTRODUCTION

The picture. An observer looks at a periodic pattern through a lens of finite resolution σ. The finer the pattern, the less of it survives the blur. Paper F [1] measured how much "information about the direction of time" a window keeps, through the quantity g. It showed that the smallest gap $g_{\min}$ falls like $1/\sqrt{\ell}$, $\ell = \ln(1/\sigma)$, with the coefficient one, for windows with at most twenty harmonics or with decay exponent $K \geq 0.003$ (which includes every window with $x \geq 0.003$). Precisely, $g_{\min}\sqrt{\ell}$ obeys explicit two-sided bounds for $10^{-100000} \leq \sigma \leq 10^{-24}$, its limit lies between $1 - \bar{\varepsilon}$ and 1 on every class with remainder bound $\bar{\varepsilon}$, and it tends to one along an explicit comb. In the language of this paper, the coefficient one is *the exchange rate between fineness and blur.* Every factor e gained in fineness costs the same amount of brightness under the blur, and what remains is a fixed fraction $1/\sqrt{\ell}$.

What is new. Paper F needed either a bound on the number of harmonics ($N \leq 20$) or a lower bound on their damping ($K \geq 0.003$). A window with many harmonics can hide its ripple in two ways. It can spread its energy thinly over a wide band, so that the sup η of the ripple stays small. Or it can put a coherent block at low frequency, so that η is large but the energy is tiny. The second way is the reason why a conjecture we used as a working hypothesis is false (Sec. IX). That conjecture was $\eta e^{\mu/2} \leq C$, where μ measures the frequency of the ripple energy. We built explicit windows with $\log_{10}(\eta e^{\mu/2})$ up to 32.0 (App. B). The coefficient one survives on all of them; only the old tool (the η sandwich) failed.

How we prove it.

- Lemma R and Lemma A4* (Sec. III) give a lower bound of g for every state. All remainders are explicit, and the remainder a_4 of the curvature step (Lemma 23 of [1]) becomes relative to the ripple.
- Lemma S (Sec. IV) replaces the sup of a low-frequency block by its energy.
- Cell-based computer-assisted proofs (Sec. VI) show that this lower bound exceeds the target.
- Lemma BE (Sec. VII) closes the small-b edge by measuring the curvature cross term in mean square instead of in sup.

Table I lists every result with its status. Throughout we keep the distinction of [1]: proof, computer-assisted proof, derivation, numerical, conjecture.

Picture. The lens sees a fine ripple only through its blurred brightness. This paper shows that the brightness that survives does not depend on how many harmonics make up the ripple, only on how much energy they carry at each frequency.

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TABLE I. Results of this paper and their status.

result	status
Lemma R (lower bound for every state)	proof (Sec. III)
Lemma A4* (a_4 relative)	proof
Lemma H (high part and θ_τ kernels)	proof (App. A)
τ cells, factor 0.99977 on $[20, 80]$	computer-assisted (App. A)
Lemma S, Lemma S′, Corollary S	proof (Sec. IV)
x-free envelope (Lemma Z3′)	proof (Sec. V)
main reduced inequality, $\ell \le 6907.8$	computer-assisted (Sec. VI)
full Lemma R* bound, $\tau \ge 20$, $b \ge 10^4\sigma^2$, $\ell \le 960.8$	computer-assisted (Sec. VI)
Lemma BE (every state with $b < 10^4\sigma^2$)	proof (Sec. VII)
seventeen thin-block windows	computer-assisted (Sec. VI)
Theorem H1 (every state with $\tau \ge 20$)	computer-assisted, modulo (P6) (Sec. VIII)
envelope values (1.7116 at 10^{-24}, 1.0063 at 10^{-3000})	numerical
conjecture C ($\eta e^{\mu/2} \le C$)	false (App. B)
local corrections (L1)–(L3), Lemma A*	proof (Sec. IX A)
local Z3′ (states with $\tau < 20$)	conjecture; ratio ≤ 2.13 on the windows tried (Sec. IX A)
full bound for $\ell > 960.8$, limit $\sigma \to 0$	open (Sec. IX B)

II. SETTING

We use the notation of [1] and recall only what is needed.

Windows and states. A periodic window is $p = \mathcal{N}_V\,\Gamma\,h/Z$. Here $\mathcal{N}_V$ is a Gaussian of variance $V = \sigma^2/b$ in v and $\Gamma = N(0, L^2)$ in w, and

$$h - 1 = e^{-w^2/2s^2} R(t), \qquad t = k'v, \tag{1}$$

$$R(t) = \sum_{n\ge1} q0_n \cos(nt - \varphi_n). \tag{2}$$

The blurred coefficients are $q0_n = |a_n| e^{-n^2K/2}$ with $K \ge x$, $x = (k'\sigma)^2$. We write

$$\eta = \sup|R|, \quad S_0 = \sum q0_n^2, \quad S_2 = \sum n^2 q0_n^2, \tag{3}$$

together with $\mu = xS_2/S_0$, $P = xS_2$, $\tau = x/b$, and $\bar{Q} = 1/4$ for the bound of (P3). A state is a choice of (b, L^2, s^2) subject to (P0) $L^2 \le 1/\sigma$ and (P4) $\sigma^2 \le s^2 \le L^2/4$. The conditions (P0)–(P7) are those of Table 5 of [1].

The quantity g. As in Eq. (3) of [1], $g \ge (1-|\varepsilon|)\,((a)+(a'))$, with

$$(a) = \frac{2A_w}{\sigma\,\mathrm{den}}, \qquad (a') = \frac{N_\phi}{\sigma\,\mathrm{den}}, \tag{4}$$

where $\mathrm{den} = \sigma^2(J_{vv} + J_{ww})$ and $|\varepsilon| \le \bar{\varepsilon}$ is the window-class constant of (P6). The statement "$g\sqrt{\ell} \ge 1$" always means "up to the factor $1 - \bar{\varepsilon}$".

Regime. This paper works in $x \le 0.003$ (the case $x \ge 0.003$ is Theorem 48 and Corollary 49 of [1]) and $x \ge 5 \times 10^{11}\sigma^2$, so that $\beta \le 10^{-6}\tau$. The states have $\tau \ge 20$, i.e. the v-focus spans at least about 0.7 ripple periods (for $\tau \ge 80$ the remainders of Lemma 23 of [1] are already below 10^{-14}). The states with $b < 10^4\sigma^2$ (the small-b edge) are treated separately in Sec. VII; there $\tau \ge 5\times10^7$ automatically. The states with $\tau < 20$ are not covered (Sec. IX A).

Picture. b is how sharply the observer focuses along v, L how wide she looks along w, and s how wide the ripple is in w. τ counts how many ripple periods fit into the focus.

III. THE LOWER BOUND FOR EVERY STATE

Lemma 1 (Lemma R)**.** *[proof] For every state, $(a) + (a') \ge \mathrm{Num}_{\mathrm{lo}}/(\sigma\,\mathrm{den}_{\mathrm{up}})$. The bounds are built from Lemmas 19, 21, 22 and 23 of [1] (called BL0, BL1, BL2 and BL3 in its App. D) and four steps:*

- *(R1) the η sandwich for J_{per} and W_{per};*
- *(R2) the upper bound of* den*, from Lemmas 21 and 22 of [1] with (R1);*
- *(R3) a bound on the correction of A_w, using $\phi \le (v'^2 + R^2)/2$ and $R^2 \le 2w^2 + 4\chi^2$;*
- *(R4) the term $e\sigma^2\langle\theta_\tau F(1-\cos\psi)\rangle$ that Lemma 23 of [1] drops, kept as the main part of (a').*

The explicit form is in `ca_reduction.py`. At the hardest band points it agrees with the reduced formula of the earlier stage (`by_eval.py`, built from the lemmas of App. D of [1]) to 4.9×10^{-12} [numerical].

The remainder a_4. In Lemma 23 of [1] the F-term of the exact identity is split as

$$\sigma^2\big(G[F] - \mathrm{Re}\,A[Fe^{i\psi}]\big) = (1-e)\sigma^2\langle\theta_\tau F\rangle + e\sigma^2\langle\theta_\tau F(1-\cos\psi)\rangle + e\sigma^2\,\mathrm{Re}\langle(\theta_\tau - \Theta)Fe^{i\psi}\rangle. \tag{5}$$

Paper F bounds the last term by $\sup F$ times the norm of $\theta_\tau - \Theta$. This gives

$$a_4 = \frac{\sigma\tau\sqrt{x}\,N_1^2\nu_1}{\sqrt{2}(1-\bar{Q})}, \tag{6}$$

an absolute term of size $e^{-\tau/2}/k'$. It is harmless for the windows of [1], where the number of harmonics or their

damping keeps N_1 bounded and the remainders are certified directly. For windows with many harmonics and $\sigma \le 10^{-48}$ it makes the background part of the bound negative at $\tau = 20$.

Lemma 2 (Lemma A4*). *[proof] With $\beta^2 = \tau y$ and $\bar\chi(w)$ the bound of $\sup_t |\chi|$ of Lemma 23 of [1],*

$$e\sigma^2 \Big| \int \Gamma \,\mathrm{Re}\langle (\theta_\tau - \Theta) F e^{i\psi} \rangle \Big| \le a_4^* = e\, W_{\mathrm{per,up}} \Big[\frac{\beta^2 \nu_2}{2} + \beta\nu_1 \Big(\frac{\xi+\zeta}{\sqrt{2}} + \sigma^2 \sqrt{\frac{s_{e2}}{\pi}}\, \frac{1}{L^2} \Big) \Big]. \quad (7)$$

Proof. Fix w, put $g = Fe^{i\psi}$, and let $\hat g_n$ be its Fourier coefficients. Pairing n and $-n$ gives

$$\mathrm{Re}\langle (\theta_\tau - \Theta) g \rangle = \sum_{n\ge 1} e_n \,\mathrm{Re}\big[(\hat g_n + \hat g_{-n})(1 - \cosh n\beta) + (\hat g_n - \hat g_{-n}) \sinh n\beta \big]. \quad (8)$$

Since $F \ge 0$, $|\hat g_n| \le \langle F \rangle$. With $\cosh u - 1 \le (u^2/2)e^u$, the first sum is at most $(\beta^2\nu_2/2)\langle F\rangle$. In the second sum write $e^{i\chi} = 1 + (e^{i\chi} - 1)$. The part with 1 is $-2\sin(w/\sqrt{2})\,\mathrm{Im}\hat F_n \sinh n\beta$. F is even in w, so its integral against the centred Γ vanishes exactly; this is the parity argument used in the proof of Lemma 23 of [1] for a_{10}. The details are in App. D 3. The rest is at most $\beta\nu_1 \sup_t |\chi|\, \langle F\rangle$, because $\sinh u \le u e^u$. Finally integrate against Γ; note $\sigma^2 \int \Gamma \langle F\rangle = W_{\mathrm{per}}$. □

a_4^* is proportional to the ripple and moves to the ripple part of the bound. Relative to the coefficient $1 - e$ of W it is of order $\tau\nu_2$ and $2\nu_1 U\sqrt{x}$; both are τ factors. On the band windows forced to $\tau = 20$, the background part changes as follows [numerical]:

σ	$T_A\sqrt{\ell}$ with a_4	$T_A\sqrt{\ell}$ with a_4^*
10^{-48}	-2.6×10^{26}	2.7×10^{23}
10^{-300}	-8.8×10^{213}	1.2×10^{149}

The ripple part stays at $T_B\sqrt{\ell} \ge 1.054$. We call Lemma R with a_4 replaced by a_4^* *Lemma R**.

Picture. The old bound counted the ripple's worst point at full weight even where the observer's focus averages it away. The new bound lets the average act first.

IV. SPLIT SANDWICHES

The η sandwich of [1] (Table 2 and App. A there) reads

$$\frac{\langle Y^2\rangle}{1+\eta} \le \Big\langle \frac{Y^2}{h} \Big\rangle \le \frac{\langle Y^2\rangle}{1-\eta} \quad (9)$$

for $Y = h_w$ and $Y = h_t$. It costs a factor $(1-\eta)/(1+\eta)$ in the bound. A coherent low-frequency block can make η large while carrying almost no energy, and then this factor is far too pessimistic.

Split $R = R_{\mathrm{lo}} + R_{\mathrm{hi}}$, where R_{lo} has frequencies in $[1, M]$ and R_{hi} frequencies $\ge m$. Let $Y = \phi(w)y(t)$ with $y = R$ or $y = R'$, and $h = 1 + \psi(w)R$ with $0 \le \psi \le 1$. Let $\langle\cdot\rangle$ be the period average integrated against a positive measure in w. Put

$$f_{\mathrm{lo}} = \frac{\|Y_{\mathrm{lo}}\|^2}{\|Y\|^2}, \quad \eta_{\mathrm{lo}} = \sup|R_{\mathrm{lo}}|, \quad \eta_{\mathrm{hi}} = \sup|R_{\mathrm{hi}}|, \quad (10)$$

$$\kappa = \min\Big(\eta_{\mathrm{lo}},\ \frac{\sup|y_{\mathrm{hi}}|\,\|R_{\mathrm{lo}}\|_t}{\|y_{\mathrm{hi}}\|_t} \Big), \quad (11)$$

$$E = \eta_{\mathrm{lo}} f_{\mathrm{lo}} + 2\eta_{\mathrm{hi}}\sqrt{f_{\mathrm{lo}} f_{\mathrm{hi}}} + (\eta_{\mathrm{hi}} + \kappa) f_{\mathrm{hi}}, \quad (12)$$

$$B = \big(\eta_{\mathrm{lo}}\sqrt{f_{\mathrm{lo}}} + \kappa\sqrt{f_{\mathrm{hi}}} + \eta_{\mathrm{hi}} \big)^2. \quad (13)$$

Lemma 3 (Lemma S). *[proof] If $2M < m$ (a spectral gap), then*

$$\|Y\|^2 \max\Big(\frac{1}{1+\eta}, 1 - E \Big) \le \Big\langle \frac{Y^2}{h} \Big\rangle \le \|Y\|^2 \min\Big(\frac{1}{1-\eta}, 1 + E + \frac{B}{1-\eta} \Big). \quad (14)$$

Proof. Use $1/(1+r) = 1 - r + r^2/(1+r)$ with $0 \le r^2/(1+r) \le r^2/(1-\eta)$, and expand $Y^2 r$. Two of the six products vanish because of the gap: $\langle Y_{\mathrm{lo}}^2 r_{\mathrm{hi}}\rangle$ and $\langle Y_{\mathrm{lo}} Y_{\mathrm{hi}} r_{\mathrm{lo}}\rangle$. The others are bounded by Cauchy–Schwarz. For the product $\langle Y_{\mathrm{hi}}^2 r_{\mathrm{lo}}\rangle$ use $\langle y_{\mathrm{hi}}^2 |R_{\mathrm{lo}}|\rangle_t \le \sup|y_{\mathrm{hi}}|\,\|y_{\mathrm{hi}}\|_t \|R_{\mathrm{lo}}\|_t$. This is where the energy $\|R_{\mathrm{lo}}\|_t^2 = S_{0,\mathrm{lo}}/2$ of the low block replaces its sup. For r^2 use Minkowski; the term $\|Y_{\mathrm{hi}} r_{\mathrm{lo}}\|$ is bounded with $\|y_{\mathrm{hi}} R_{\mathrm{lo}}\|_t \le \sup|y_{\mathrm{hi}}|\,\|R_{\mathrm{lo}}\|_t$ (or by η_{lo}), so the same κ enters B as E. The line-by-line proof, with the norms and measures, is in App. D 2. □

Lemma 4 (Lemma S′, no gap). *[proof] Without the gap, Lemma 3 holds with E replaced by*

$$E' = E + \eta_{\mathrm{hi}} f_{\mathrm{lo}} + 2\eta_{\mathrm{lo}}\sqrt{f_{\mathrm{lo}} f_{\mathrm{hi}}}. \quad (15)$$

Proof. The two products that used the gap are bounded by $\eta_{\mathrm{hi}}\|Y_{\mathrm{lo}}\|^2$ and $\eta_{\mathrm{lo}}\|Y_{\mathrm{lo}}\|\|Y_{\mathrm{hi}}\|$. The upper bound B used Minkowski only, so it is unchanged. □

Scaling. Let $\lambda = \min(\mathrm{low}_{S_0}, \mathrm{low}_{S_2})$ and $\Upsilon = \mathrm{up}_{S_2}$ be the lower and upper factors of the two channels. The part of den that carries S_0 enters the target only through $\gamma\sigma$. In the cells of Sec. VI it is given its factor $1/(1-\bar\eta) = 2$ directly. Then a certificate made with the factors set to one gives $g \ge (\lambda/\Upsilon)\gamma$.

Corollary 5 (Corollary S). *[proof] If the certified target is $T_{10} = 1 + 12.1/\ell + 2.5 \times 10^{-4}$ and $(\lambda/\Upsilon)T_{10} \ge 1$, then $g\sqrt{\ell} \ge 1$ (times $1 - \bar\varepsilon$).*

Checks. On the nine windows that refute conjecture C, $(\lambda/\Upsilon)T_{10}$ lies between 1.0020 and 1.1074 [numerical]. These are exact sums over the window. A search for counterexamples to Lemmas 3 and 4 is described in App. B. It covered low and high frequencies, few and many harmonics, coherent and random phases, and used a hill climb. It found no violation. The same search does find violations of deliberately wrong versions in which $\eta_{\rm lo} f_{\rm lo}$, $\eta_{\rm hi} f_{\rm hi}$ or B is dropped. It finds none when the term $2\eta_{\rm hi}\sqrt{f_{\rm lo} f_{\rm hi}}$, the term $\kappa f_{\rm hi}$ or the gap-free cross term is dropped. Whether those three terms are needed is therefore not settled.

Picture. A low-frequency block is like a slow swell under the fine ripple. It can be tall, but the lens reads the ripple's detail through its energy, and a slow swell carries almost none of the detail.

V. THE X-FREE REDUCED INEQUALITY

The main reduced function. Put $A = \sigma^2/s^2$, $r = s^2/L^2 \in (0, 1/4]$, $g_s = \sqrt{r/(2+r)}$ and $G = P\, g_s$. The leading terms of Lemma R are

$$N = 2Ar + \frac{GA}{\mu(2+r)} + \frac{3\sigma^2}{8} + \frac{b\sigma^2}{8rA} + \frac{G\sigma^2}{16b} + \frac{G\sigma^2}{16A(2+r)}, \quad (16)$$

$$D = b + \frac{G}{2} + Ar + \frac{GA}{2\mu(2+r)}. \quad (17)$$

Here $g \geq \gamma$ holds if $N - \gamma\sigma D \geq 0$. N is of the form $\alpha A + \beta_1/A + \cdots$, so A can be eliminated exactly by the inequality of arithmetic and geometric means. The result is a function $\phi(\mu, r, b, G)$; $\phi \geq 0$ implies $g \geq \gamma$. For fixed (μ, r, b) it is concave in G, so the two end points $G = 0$ and $G = G_{\max}$ suffice.

Lemma 6 (Lemma Z3′, x-free form)**.** *[proof] For $x \leq 0.003$ and $\mu \geq 2$, $P \leq \Pi^*(\mu)$, where*

$$\Pi^*(\mu) = \bar{Q}^2 \mu\, \frac{1 + a t_0}{(1+t_0)^2}, \quad (18)$$

with

$$t_0 = \frac{e}{2}(\mu - 0.003)\, e^{(\mu - 0.003)/2}, \qquad a = c_s e^{-\mu/2}, \quad (19)$$

and $c_s = e^{0.0015}(1 + \sqrt{e} + 2/e) = 3.3896$ (the value at $x = 0.003$). For every $\mu > 0$, $P \leq \mu\bar{Q}^2$.

The proof is the fractional knapsack argument of the earlier stages, with x removed at its largest value; it is written out in App. D 4. The monotonicity step needs $a \leq 2$, i.e. $\mu \geq 1.055$; the cells use Π^* only for $\mu \geq 2$ and $\mu\bar{Q}^2$ below. Nothing is lost: the x-free envelope reproduces the envelope minimised over x. For example $g\sqrt{\ell} = 1.7116$ at $\sigma = 10^{-24}$, 1.3109 at 10^{-48}, 1.1508 at 10^{-96}, 1.0509 at 10^{-300} and 1.0063 at 10^{-3000} [numerical]. The envelope approaches one from above, roughly like $1 + 40/\ell$.

Picture. Of all ripples with the same energy frequency, the hardest one is the one that hides as much energy as the knapsack allows just below the blur. The envelope is the brightness that even this ripple must leave.

VI. COMPUTER-ASSISTED PROOFS

All cell proofs share one design. The parameters are cut into boxes in $(\log\mu, \log r, \log b)$. On each box every term is bounded by its worst corner; each term is monotone in each variable. Terms with the same power of σ are combined at equal σ before the worst end of the ℓ cell is taken. Everything is computed in logarithms, and underflow may only drop non-negative parts. A box is accepted only when its margin exceeds a derived bound of the rounding error times the safety factor $S = 10$; otherwise it is bisected. The floating-point model, the bounds and the rule are stated in App. E. Tails in μ and b are separate boxes.

(i) The main reduced inequality. With $\gamma = T_D(\ell)/\sqrt{\ell}$ and $T_D = 1 + (2.1 + D)/\ell + 2.5 \times 10^{-4}$:

- for $D = 0$, the 301 columns $\ell \in [55.26, 3062.6]$ close, with 465900 audit points and no violation;
- for $D = 10$, all 686 columns $\ell \in [55.26, 6907.8]$ close ($\sigma = 10^{-24}$ to 10^{-3000}), with 1111902 audit points and no violation.

A column needs up to 7.8×10^7 boxes. Every accepted box has $\hat{\ell}_{\rm pos} - \hat{\ell}_{\rm neg} \geq S\delta$ (App. E); the smallest ratio $(\hat{\ell}_{\rm pos} - \hat{\ell}_{\rm neg})/(S\delta)$ over all accepted boxes is 1.00, as it must be, since a box is bisected until the rule holds. [proof, computer-assisted: `cd_cells.py` with the rule of `cl_cells.py`] The certificate is not vacuous: at $\ell = 55.26$ the target 1.70 (below the envelope 1.711) closes, and 1.73 does not.

(ii) The full Lemma R bound.* Here all remainders of Lemma R* are kept:

- the corrections of A_w;
- the saturation of Ψ and Φ;
- both forms of the F remainder;
- a_4^*;
- a_1–a_{10} without a_4;
- the ν factors at $\tau = 20$.

The window enters only through generic bounds: $\sqrt{x} N_1 \leq \bar{Q}\sqrt{2\ell + 3.1} + 10^{-6}$, $x N_2 \leq \bar{Q}(2\ell + 3.1) + 10^{-4}$, and $\bar{\eta} = 1/2$ in the remainders. For $\tau \geq 20$ and $b \geq 10^4\sigma^2$, the 92 columns $\ell \in [55.26, 960.8]$ close with target

T_{10} (σ down to about 10^{-417}). Every term is moved by $S\eta_t + 1000Su$ in the safe direction before the terms are netted (App. E), so that the cancellation between equal powers of σ is covered. The states with $b < 10^4\sigma^2$ are covered by Lemma 7 (Sec. VII). [proof, computer-assisted: `cl_link.py`] Pointwise, the bound agrees with the main reduced function to about 2×10^{-4} near the envelope. Its soundness against a decimal evaluation of Lemma R* was checked at random states [numerical].

(iii) Seventeen thin-block windows. For the intermediate windows not covered by Corollary 5, the Lemma S factors are put inside the boxes. For each window, μ, x and P are fixed at the window's values, and $G \le Pg_s(r)$ is used exactly. All 17 windows close for the target $g\sqrt{\ell} \ge 1$, with the rule of (ii). [proof, computer-assisted: `cf_windows.py with cl_link.py`] For one of them we checked that raising the target to 1.2 times the window's upper value makes the proof fail, as it must.

Picture. The computer does not search for the best state. It checks that no state, in any box, can do worse than the target, and every box keeps a cushion ten times larger than the largest possible rounding error.

VII. THE SMALL-b EDGE

The certificates of Sec. VI cover $b \ge 10^4\sigma^2$. Below this edge the cells cannot work with the generic bound γ_P of the ripple derivative: the term $-\sigma^2\gamma_P\Phi$ of the curvature step is not tied to the size of the ripple. We close the edge by measuring the cross term of Lemma 23 of [1] (BL3) in mean square instead of in sup. The coefficient need not be one here; the bound is of order $1/\sigma$.

Lemma 7 (Lemma BE)**.** *[proof] Let $\sigma \in [10^{-3000}, 10^{-24}]$, $x \in [5 \times 10^{11}\sigma^2, 0.003]$, (P0), (P3), (P4), and $q0_n \le 2e^{-n^2x/2}$. Then every state with $0 < b \le 10^4\sigma^2$ satisfies*

$$(a) + (a') \;\ge\; \frac{\kappa}{\sigma}, \qquad \kappa = 1.1 \times 10^{-5}. \tag{20}$$

Hence $g\sqrt{\ell} \ge (1-\bar\varepsilon)\,\kappa\sqrt{\ell}/\sigma \ge (1-\bar\varepsilon)\,8 \times 10^{19}$. Condition (P2) is not used, and there is no restriction on x inside the regime. The bound on $q0_n$ holds for every window of [1]: a positive profile has $|a_n| \le 2$, and $q0_n = |a_n|e^{-n^2K/2}$ with $K \ge x$ (Sec. II).

The proof has four steps. Write $E = e^{-w^2/2s^2}$, so that $h = 1 + ER$.

(BE-a) The ripple derivative. Let $G = \partial_v(\tilde s + s_w)$ as in App. A of [1]. A direct computation, using $h - ER = 1$, gives

$$\sigma^2 G = E\Big[\frac{xR''}{h} - \frac{xER'^2}{h^2} - \sigma\sqrt{x}\,\frac{w}{s^2}\,\frac{R'}{h^2}\Big]. \tag{21}$$

Hence $|\sigma^2 G| \le E\,[xN_2(1-\bar Q) + xEN_1^2 + \sigma\sqrt{x}\,(|w|/s^2)N_1]/(1-\bar Q)^2$. Its supremum over w is below the γ_P of [1] (Table 2 there). The envelope E shows that $\sigma^2 G$ lives where the ripple lives.

(BE-1) The cross term in mean square. By the parity identity $\langle hG\rangle = -\langle F\rangle$ (proof of Lemma 23 of [1]), for every $c > 0$ and every w,

$$\langle h\sigma^2 G\cos\psi\rangle \ge -\sigma^2\langle F\rangle - \frac{\langle h(\sigma^2 G)^2\rangle}{4c} - 2c\,\langle h(1-\cos\psi)\rangle. \tag{22}$$

Here we used $(1-\cos\psi)^2 \le 2(1-\cos\psi)$. In Lemma 23 of [1] this replaces only the step $\ge -\sigma^2\langle F\rangle - \gamma_P\langle h(1-\cos\psi)\rangle$. The remainders a_5, a_6 keep γ_P; they carry ν factors.

(BE-2) The mean square. Put $M_G = \int \Gamma\langle h(\sigma^2 G)^2\rangle$. From (BE-a), $\langle R''^2\rangle = S_4/2$, $\langle R'^2\rangle = S_2/2$, $\langle R'^4\rangle \le N_1^2 S_2/2$, $\int \Gamma E^2 = g_s$ and $\int \Gamma E^2 w^2/s^4 \le g_s/(2s^2)$,

$$M_G \le \frac{3g_s}{2}\Big[\frac{x^2 S_4}{1-\bar Q} + \frac{xS_2\,(U_1^2 + \frac12)}{(1-\bar Q)^3}\Big], \tag{23}$$

with $U_1 \ge \sqrt{x}N_1$ the window constant of Sec. VI. For any $\Lambda > 0$, $x^2S_4 \le \Lambda\, xS_2 + T_\Lambda$, where $T_\Lambda \le 4\sum_{xn^2>\Lambda}(xn^2)^2 e^{-xn^2}$.

(BE-3) Choice and comparison. Take $c = (b + \sigma^2 J_{ww})/(16\sigma^2)$ and $\Lambda = 10^{-3}/\sigma^2$. With $\Phi \le 4\Psi$ and $\Phi \le 2$, the Φ part costs at most $b\Psi/2 + \sigma^2 J_{ww}/4$. The cross term enters Lemma 23 of [1] with a factor $e(1-b)\sigma^2$, so the M_G part is $e\sigma^2 M_G/(4c) = 4e\sigma^4 M_G/(b + \sigma^2 J_{ww})$. With $xS_2 g_s \le 2(1+\bar Q)W_{\rm per}$ it is at most

$$\rho_W(1-e)W_{\rm per} + \frac{6\sigma^2 g_s T_\Lambda}{(1-\bar Q)J_{ww}}, \tag{24}$$

where

$$\rho_W = \frac{48(1+\bar Q)K_1\sigma^2}{(1-b)^2} \le 0.0801, \tag{25}$$

and $K_1 = \Lambda/(1-\bar Q) + (U_1^2 + \frac12)/(1-\bar Q)^3$. Here $y = (1-b)^2\sigma^2/(2b) \ge 4.99 \times 10^{-5}$, so $1 - e \ge y/(2+y) \ge 2.49 \times 10^{-5}$. With $2A_w \ge \sigma^2 J_{ww}$ ($\phi \le 2$) and $\sigma^2 J_{vv} = b + W$ (Lemma 22 of [1]), this gives

$$2A_w + N_\phi \ge \frac{7}{8}\sigma^2 J_{ww} + 0.45(1-e)W + \frac14 b\Psi - \text{rem}. \tag{26}$$

Every remainder (the a_i of Lemma 23 of [1] with a_4^*, the Φ-proportional terms, the ν terms of the W sandwich, and T_Λ) is below 10^{-10^7} times its comparison quantity. The reason is $\tau = x/b \ge 5 \times 10^7$. For $L^2 \ge 1$, $\Psi \ge \min(\Phi, \frac12) \ge 0.2211$. For $L^2 < 1$, $\sigma^2 J_{ww} > 0.6\sigma^2 \ge 6 \times 10^{-5} b$. In both cases $2A_w + N_\phi \ge 1.14 \times 10^{-5}$ den. Each step is written out in App. D 5. □

The constants, the formula of (BE-a) against finite differences, the bound of (BE-2) on random windows and every remainder are checked in `ch_bedge.py` [numerical].

Picture. When b is tiny the observer's v-focus is wide. Then the slope information W of the ripple enters almost undiminished. The cross term that blocked the edge was measured at the ripple's worst point. Measured in mean square, it is a few per cent of W.

VIII. MAIN THEOREM

Theorem 8 (Theorem H1)**.** *[proof, computer-assisted: Secs. VI and VII] Let a periodic window satisfy (P0), (P2)–(P4), (P6), $x \le 0.003$ and $x \ge 5\times10^{11}\sigma^2$, with $\ell \in [55.26, 960.8]$. Then every state with $\tau = x/b \ge 20$ satisfies*

$$g\sqrt{\ell} \;\ge\; (1-\bar\varepsilon)\,\frac{\lambda}{\Upsilon}\Big(1+\frac{12.1}{\ell}+2.5\times10^{-4}\Big). \qquad (27)$$

Every state with $b < 10^4\sigma^2$ satisfies the much stronger bound $g\sqrt{\ell} \ge (1-\bar\varepsilon)\,8\times10^{19}$ (Lemma 7). Here λ/Υ is the sandwich ratio of Sec. IV: $(1-\eta)/(1+\eta)$ for the η sandwich, or the Lemma S/S' factors. In particular, over the states with $\tau \ge 20$, $g_{\min}\sqrt{\ell} \ge 1-\bar\varepsilon$ holds for:

(i) every window with $\eta \le 1/\ell$;

(ii) every window with $(\lambda/\Upsilon)T_{10} \ge 1$, which includes the nine windows that refute conjecture C;

(iii) the seventeen thin-block windows of Sec. VI.

In the observer's words, the condition $\tau \ge 20$ excludes only the states whose v-focus is narrower than about 0.7 ripple periods (precisely, $k'\sqrt{V} < \sqrt{20}$). These states are the subject of Sec. IX.

Proof. Split the states with $\tau \ge 20$ at $b = 10^4\sigma^2$.

States with $b \ge 10^4\sigma^2$. Lemma 1 with Lemma 2 gives $g \ge (1-\bar\varepsilon)\,\mathrm{Num}/(\sigma\,\mathrm{den})$. Certificate (ii) of Sec. VI gives $\mathrm{Num} - \gamma\sigma\,\mathrm{den} \ge 0$ with the factors set to one, for $b \in [10^4\sigma^2, 0.003/20]$. Every state with $\tau \ge 20$ has $b = x/\tau \le 0.003/20$. The scaling argument of Sec. IV restores the factors.

States with $b < 10^4\sigma^2$. Lemma 7 gives $(a)+(a') \ge 1.1\times10^{-5}/\sigma$, which is above γ by many orders of magnitude. The sandwich ratio is at most one, so the bound of the theorem follows.

For (i), $(1-1/\ell)(1+12.1/\ell+2.5\times10^{-4})/(1+1/\ell) \ge 1$ on the whole range. For (ii) see Corollary 5. For (iii) see certificate (iii). □

Relation to [1]. Theorem 8 does not need a bound on the number of harmonics. Paper F covers windows with at most twenty harmonics or with damping exponent $K \ge 0.003$ (Theorem 12 and Corollary 13 of [1], with Theorems 45 and 48 for the two classes), for $10^{-100000} \le \sigma \le 10^{-24}$ and in the limit, with both sides of the law. Theorem 8 covers the lower side for windows with a weak or energy-poor ripple, for σ from 10^{-24} down to about 10^{-417} and for every state with $\tau \ge 20$. The main reduced inequality alone (certificate (i)) holds up to $\ell = 6907.8$.

Picture. However the ripple is built, many harmonics or few, the observer keeps at least the fraction $1/\sqrt{\ell}$, as long as the ripple's sup or its low-frequency energy is small and her focus spans at least about 0.7 ripple periods.

IX. THE ROAD TO AN UNCONDITIONAL THEOREM

Theorem 8 has five restrictions. We list them from the nearest to the farthest, each with what is known and its status. The small-b edge is no longer among them: Lemma 7 closes it.

A. States with a narrow focus ($\tau < 20$)

What changes. When τ is small the observer sees the ripple only near the focus position t_0, through the wrapped normal θ_τ (Lemma 19 of [1]). Lemma R replaces these local averages by period averages and pays with factors $\nu_k \sim e^{-\tau/2}$. This route has a ceiling: the τ factor of the ripple part, $\sqrt{(1-\epsilon_1)(1-\nu_0)/(1+z)}\,(1-z)/(1+\nu_0)$, falls below T_{10}/env at $\tau \approx 5.4$ ($\ell = 55.26$), 10.1 ($\ell = 690.8$) and 14.1 ($\ell = 6907.8$) [derivation]. Below the ceiling, neighbouring high harmonics beat: their difference frequency is not averaged out by a narrow focus. For two neighbours $m = 50, 51$ the focused slope energy $\langle\theta_\tau R'^2\rangle/\langle R'^2\rangle$ ranges over $[0.005, 1.995]$ at $\tau = 0.01$, although $m^2\tau = 25$ [numerical].

Local averages [derivation]. Write $S_{0,\theta} = 2\langle\theta_\tau R^2\rangle$, $S_{2,\theta} = 2\langle\theta_\tau R'^2\rangle$, $\mu_\theta = xS_{2,\theta}/S_{0,\theta}$ and $P_\theta = xS_{2,\theta}$. Lemma R can be rewritten with these local quantities and with $W_\theta = x\int\Gamma\langle\theta_\tau h_t^2/h\rangle$, $J_\theta = \int\Gamma\langle\theta_\tau h_w^2/h\rangle$, without ever replacing θ_τ by one. Its ripple part keeps the algebraic form of Lemma R, and the factors ν_k of the ripple part disappear. Three local corrections appear, each proved [proof]:

- (L1) the focus term of the w-information of the background is at most $2\bar Q\sqrt{r/(1+r)}/L^2$, $r = s^2/L^2$;
- (L2) the curvature term of W is at most $(2/e)\bar Q\,b = 0.184\,b$ for every τ, because $xn^2e^{-n^2\tau/2} = b\,u\,e^{-u/2} \le 2b/e$ with $u = n^2\tau$;
- (L3) the normalisation obeys $|Z-1| \le \bar Q\sqrt{r/(1+r)}$.

The remaining remainder is $\Theta - \theta_\tau$. By Lemma A* below it is at most $\beta\nu_1\langle|g|\rangle$, with $\beta\nu_1 \le 2\times10^{-6}$. Its right side is a period average; comparing it with the local quantities

at a node of the focused ripple and at very small τ is still to be done. A direction we have not followed: bound $\Theta - \theta_\tau$ in real space, through the Gaussian integral in v itself, with $|e^{-\alpha^2 V/2} - e^{i\alpha v}| \le \alpha^2 V/2 + \alpha|v|$; the right side is then a local average, although with a Gaussian weight of a different width. Harmonics with $n^2\tau < 20$ carry the energy $x n^2 q0_n^2 < 20b\, q0_n^2$, at most $1.25\,b$ in total [proof]; they are of the size of the background. A second direction: combine the local quantities with Lemma S, measuring the low part of the focused ripple by its energy rather than its sup, as Lemma S does for period averages [conjecture].

The core: a local Z3′ [conjecture]. With the local form of Lemma R, everything else would follow from

$$P_\theta \le C\,\Pi^*(\mu_\theta) \tag{28}$$

for a universal constant C. In words: even inside a narrow focus, a finer ripple must be darker. The reason to expect it is non-negativity: the raw profile is $f \ge 0$, so a coherent wave packet in it has height at most one, and the blur damps frequency n by $e^{-n^2 x/2}$. On windows satisfying (P2) (beats, wave packets, packets placed at a zero of the fundamental, combs, random bands; focus positions on a grid of 512, τ from 20 down to $2/n^2$), the largest ratio $P_\theta/\Pi^*(\mu_\theta)$ we found was 2.13 [numerical]. It occurs at the crest of the fundamental, where $\sin^2$ is twice its mean. The coefficient tolerates much more. With the main terms of the local form, (L1)–(L3) at their worst and the brightness raised to $C = 10$, the envelope stays above T_{10} at every ℓ we computed, for example 1.4664 at $\ell = 55.26$ and 1.0103 at $\ell = 3453.88$ [numerical].

Lemma A [proof].* In Lemma 23 of [1], every remainder of the form $\nu \sup_t |g|$ may use $\nu\langle|g|\rangle$ instead. The kernels $\Theta - 1$, $\theta_\tau - \Theta$ and $\partial_t\Theta$ have absolutely summable Fourier series with the same sums, and $|\hat g_n| \le \langle|g|\rangle$. With $\langle h|\sigma^2 G|\rangle \le \langle h(\sigma^2 G)^2\rangle^{1/2}$ and (BE-2), the remainders a_5 and a_6 become proportional to the ripple. In the cells of certificate (ii) this lowers the reachable τ. Single columns close for $\tau \ge 8$ at $\ell \in [55.26, 65.26]$, for $\tau \ge 10$ at $\ell \in [215.26, 225.26]$, and for $\tau \ge 12$ at $\ell \in [455.26, 465.26]$, $[695.26, 705.26]$ and $[855.26, 865.26]$ [proof, computer-assisted: `ci_link.py`, single columns]. A full sweep for $\tau \ge 12$ is in progress and is not used in Theorem 8.

B. The range of σ

The full bound is certified for $\ell \le 960.8$ (certificate (ii)); a column near $\ell = 950$ needs 8.2×10^7 boxes. The main reduced inequality alone is certified up to $\ell = 6907.8$ (certificate (i)). A relative bound of the remainders by the main numerator would carry the full bound to that range; the available margin is $(T_{10} - T_0)/T_{10} \ge 1.4 \times 10^{-3}$. Point searches indicate that beyond $\ell \approx 3500$ the old remainder a_6 needs $\tau \ge 30$ [numerical]; Lemma A* removes the window-generic γ_P from a_6. The limit $\sigma \to 0$ cannot be reached by finitely many cells; it needs an analytic lemma for large ℓ, as in Corollary 13 of [1].

C. Which windows have coefficient one

For a general window the theorem keeps the factor λ/Υ. Coefficient one is proved for the windows (i)–(iii) of Theorem 8. A possible direction [conjecture]: if λ/Υ could be bounded through the concentration of the spectrum, for example through the cumulative share $F_N = \sum_{n\le N} n^2 q0_n^2/S_2$, the seventeen window-by-window certificates might be replaced by one sufficient condition.

Conjecture C is false. In earlier work we used the working hypothesis $\eta e^{\mu/2} \le C$, with $C \approx 1$. Explicit windows refute it (App. B). Each is a Rudin–Shapiro band plus a wide coherent block of about 10^8 to 10^{70} harmonics with a tiny raw amplitude. They reach $\log_{10}(\eta e^{\mu/2}) = 32.0$ at $\sigma = 10^{-300}$. The reason is that η adds up coefficients (an ℓ^1 quantity) while μ is set by energies (an ℓ^2 quantity). The weak form $\mu \le 2\ln(1/\eta) + O(\ln\ln)$ is false as well. The coefficient one is not affected: the upper value of $g\sqrt{\ell}$ on these windows is 1.12 to 2.14, above the band alone, and Theorem 8(ii) covers them.

D. Hypotheses on the window class

- The constant $\bar\varepsilon$ of (P6) is not closed for band windows; the third-order terms are open [conjecture].
- (P2) is used through Lemma Z3′. Without (P2) the bound fails already for period averages: windows without a dominant fundamental (beats, combs) exceed $\Pi^*(\mu)$ by factors 3.48 to 318 [numerical]. Whether a weaker hypothesis suffices is open.
- $x \ge 5\times 10^{11}\sigma^2$ makes $\beta \le 10^{-6}\tau$ and gives $\tau \ge 5\times 10^7$ on the small-b edge; it is a technical condition.

E. The upper side

This paper proves only the lower side. The upper side ($g\sqrt{\ell} \le 1 + \dots$) for general windows has not been attempted; for the windows of [1] it is proved there (Theorem 12 and Corollary 13 of [1]).

Picture. The nearest restriction is the narrow focus. There the observer sees the ripple's brightness locally, and the one missing inequality says that local brightness, too, must pay for fineness.

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The author used AI assistants in this work. Anthropic Claude (claude-opus-5-5) was used in two separate sessions. In the first session it planned the study and wrote

the instructions for each stage, re-ran every verification suite and re-checked a column of the certificates independently, pointed out that the main theorem had to cover every state of a window (and not only the states near the minimum of the envelope) before it could be stated for $g_{\min}$, found that a sentence of the draft misstated the scope of paper F, and reviewed and edited the manuscript. In the second session it derived the lemmas (Lemma R, Lemma A4*, Lemma H, the split sandwich, the x-free envelope, Lemma BE and the local corrections of Sec. IX A), wrote the cell-based computer-assisted certificates, constructed the counterexamples to conjecture C and the thin-block windows, wrote the evaluation and verification code, and drafted the text; it also found and reported weaknesses of its own and of the first session's reasoning, among them that a set of numbers used as evidence for the narrow-focus states was $\sqrt{\ell}$ itself, and corrected the rounding direction of several constants. While writing the specification of the cells, the second session found that the original acceptance rule did not by itself cover floating-point rounding at large ℓ; it derived the rounding bounds of App. E and recomputed every certificate with the new rule, and all of them closed again. Results were passed between the sessions as written summaries and checked in the first session. Google Gemini (Gemini 4 Argon) and OpenAI ChatGPT (GPT-5.6 Luna) independently reviewed the manuscript, re-deriving the main steps of Lemmas A4*, S, S′ and BE; their comments led to the line-by-line proofs of App. D, the specification of the computer-assisted proofs in App. E, the restriction of Lemma Z3′ to $\mu \ge 2$, references to paper F by its published numbering, and several corrections of notation. The author directed the research, made all decisions on its content, and takes full responsibility for this article.

Data and code availability. All scripts, results and the verification suite (`verification/run_all.py`) will be released with this paper. Paper F and its repository [1] are not changed.

Appendix A: The τ factors and Lemma H

Lemma 9 (Lemma H)**.** *[proof]Split $h - 1 = e^{-w^2/2s^2}(R_{\rm lo} + R_{\rm hi})$ at frequency m. For the kernels $K = 2\sum \kappa_n \cos nt$ of App. D of [1],*

$$|\langle K(h-1)\rangle| \le \eta_{\rm lo}\|K\|_2 + \eta_{\rm hi}\|K_{\ge m}\|_2. \qquad \text{(A1)}$$

Since $\kappa_n \propto n^k e^{-n^2\tau/2+n\beta}$, the high part carries $e^{-m^2\tau/2}$, and $m^2\tau = u^2/b \ge u^2$.

τ cells. [proof, computer-assisted: `cb_tau.py`] The τ-dependent factors of the bound decrease in τ and increase in β. On 400 cells of $[20, 80]$, with $\beta \le 10^{-6}\tau$ and an audit of 20000 points, their product is at least 0.99977. On $[40, 80]$ it is at least $1 - 1.1 \times 10^{-8}$. In certificate (ii) of Sec. VI the ν's are used directly at $\tau = 20$. There $\tau^j \nu_k(\tau, 10^{-6}\tau)$ decreases on $[20, 400]$ (checked).

Appendix B: Windows

Rudin–Shapiro bands. $f = 1 + c\cos t + c\sum_{k<W} \epsilon_k \cos((m+k)t)$ with the Rudin–Shapiro signs ϵ_k, $W = 2^j$. Non-negativity follows from $|P_j|^2 + |Q_j|^2 = 2W$. The blurred sup is bounded by Abel summation with the partial sums $(2 + \sqrt{2})\sqrt{n}$.

Counterexamples to conjecture C. Add a coherent block (all phases zero) of N_c harmonics at $u_c = 0.3$ with raw amplitude A. Non-negativity follows from $1 - c(1 + \sqrt{2W}) - A(1 + N_c) \ge 0$; (P2) and (P3) hold by construction.

σ	η	μ	$\log_{10} \eta e^{\mu/2}$	upper $g\sqrt{\ell}$
10^{-48}	0.027	15.15	1.7	1.492
10^{-96}	0.071	38.08	7.1	1.423
10^{-300}	0.027	154.8	32.0	1.120
10^{-300}	0.071	149.6	31.3	1.248
10^{-300}	0.111	134.7	28.3	1.427

Search for counterexamples to Lemma S. The search draws gapped and gap-free windows with 1 to 300 harmonics on each side, coherent or random, and hill-climbs on the worst margin. We used 300 windows of each kind; the smallest margins were 6.7×10^{-3} and 1.3×10^{-2}.

Appendix C: Verification

`verification/run_all.py` recomputes a sample of every certificate in less than thirty minutes; `--recompute` recomputes all of them. It also re-derives every number of the manuscript from `results/` and checks that it appears in the text. Lemma BE is checked by `ch_bedge.py`; the local quantities of Sec. IX A by `ci_rtheta.py`, `ci_local.py` and `ci_envloc.py`; the single columns with Lemma A* by `ci_link.py`. The acceptance rules of App. E are implemented in `cl_cells.py` and `cl_link.py`; the bounds are checked against long double by `ck_rounding.py`, and the τ cells and Lemma BE by `cl_tau_be.py`. Each step of App. D is checked independently, in exact fractions and 60-digit decimal, by `ck_details.py`. The stage scripts are listed with their outputs in the repository README.

Appendix D: Detailed proofs

This appendix writes the new lemmas out line by line. Each step has an independent check in `verification/ck_details.py` (exact rational arithmetic, Fourier sums, random states, and 60-digit decimal constants); it runs in the default of `run_all.py`.

1. Conventions

Functions are 2π-periodic in t and depend on w. For such a function $u(t,w)$ and a positive measure m on the w-axis we write

$$\langle u\rangle_t = \frac{1}{2\pi}\int_0^{2\pi} u\,dt, \qquad \langle u\rangle = \int \langle u\rangle_t\, dm(w), \tag{D1}$$

$$\|y\|_t^2 = \langle y^2\rangle_t, \qquad \|u\|^2 = \langle u^2\rangle. \tag{D2}$$

So $\langle\cdot\rangle_t$ is the normalised period average, and $\|\cdot\|$ is the L^2 norm of $dt/2\pi\otimes m$. Fourier coefficients are $\hat u_n = \langle u\,e^{-int}\rangle_t$, so $u=\sum_n \hat u_n e^{int}$ and $\langle KU\rangle_t = \sum_n \hat K_n \hat U_{-n}$. For $R=\sum_{n\ge1} q0_n\cos(nt-\varphi_n)$, Parseval gives $\|R\|_t^2 = S_0/2$ and $\|R'\|_t^2 = S_2/2$.

2. Lemma S

Data. $Y=\phi(w)y(t)$ with $y=R$ or $y=R'$, $h=1+r$ with $r=\psi(w)R(t)$, $0\le\psi\le1$, $\eta=\sup|R|<1$. The split $R=R_{\rm lo}+R_{\rm hi}$ puts the frequencies $\le M$ in $R_{\rm lo}$ and those $\ge m$ in $R_{\rm hi}$, $M<m$. Then $y=y_{\rm lo}+y_{\rm hi}$, $Y=Y_{\rm lo}+Y_{\rm hi}$, $r=r_{\rm lo}+r_{\rm hi}$, with $|r_{\rm lo}|\le\eta_{\rm lo}$ and $|r_{\rm hi}|\le\eta_{\rm hi}$.

Step 1 (orthogonality). $y_{\rm lo}$ and $y_{\rm hi}$ have disjoint frequencies, so $\langle y_{\rm lo}y_{\rm hi}\rangle_t=0$. Hence $\|Y\|^2=\|Y_{\rm lo}\|^2+\|Y_{\rm hi}\|^2$ and $f_{\rm lo}+f_{\rm hi}=1$.

Step 2 (algebra). For $|r|<1$,

$$\frac{1}{1+r} = 1-r+\frac{r^2}{1+r}, \qquad 0\le\frac{r^2}{1+r}\le\frac{r^2}{1-\eta}. \tag{D3}$$

Hence

$$\Big\langle\frac{Y^2}{h}\Big\rangle = \|Y\|^2 - \langle Y^2 r\rangle + \Big\langle\frac{Y^2r^2}{1+r}\Big\rangle, \tag{D4}$$

with the last term in $[0,\|Yr\|^2/(1-\eta)]$.

Step 3 (six products). Expanding,

$$\begin{aligned}\langle Y^2r\rangle = {}&\langle Y_{\rm lo}^2 r_{\rm lo}\rangle + \langle Y_{\rm lo}^2 r_{\rm hi}\rangle + 2\langle Y_{\rm lo}Y_{\rm hi}r_{\rm lo}\rangle\\ &+ 2\langle Y_{\rm lo}Y_{\rm hi}r_{\rm hi}\rangle + \langle Y_{\rm hi}^2 r_{\rm lo}\rangle + \langle Y_{\rm hi}^2 r_{\rm hi}\rangle.\end{aligned} \tag{D5}$$

(a) $|\langle Y_{\rm lo}^2 r_{\rm lo}\rangle|\le\eta_{\rm lo}\|Y_{\rm lo}\|^2$. (b) With the gap $2M<m$: $y_{\rm lo}^2$ has frequencies $\le 2M$ and $R_{\rm hi}$ frequencies $\ge m$, so $\langle y_{\rm lo}^2 R_{\rm hi}\rangle_t=0$ and $\langle Y_{\rm lo}^2 r_{\rm hi}\rangle=0$. (c) With the gap: $y_{\rm lo}R_{\rm lo}$ has frequencies $\le 2M$, $y_{\rm hi}$ frequencies $\ge m$, so $\langle Y_{\rm lo}Y_{\rm hi}r_{\rm lo}\rangle=0$. (d) By Cauchy–Schwarz in $\langle\cdot\rangle$, $|\langle Y_{\rm lo}Y_{\rm hi}r_{\rm hi}\rangle|\le\eta_{\rm hi}\|Y_{\rm lo}\|\|Y_{\rm hi}\|$. (e) Two bounds hold. First, $|\langle Y_{\rm hi}^2 r_{\rm lo}\rangle|\le\eta_{\rm lo}\|Y_{\rm hi}\|^2$. Second, with $0\le\psi\le1$ and Cauchy–Schwarz in t,

$$\begin{aligned}|\langle Y_{\rm hi}^2 r_{\rm lo}\rangle| &\le \int \phi^2\, dm\,\langle y_{\rm hi}^2|R_{\rm lo}|\rangle_t\\ &\le \int\phi^2\, dm\,\sup|y_{\rm hi}|\,\|y_{\rm hi}\|_t\|R_{\rm lo}\|_t.\end{aligned} \tag{D6}$$

Since $\|Y_{\rm hi}\|^2=\int\phi^2 dm\,\|y_{\rm hi}\|_t^2$, the right side is $\|Y_{\rm hi}\|^2\sup|y_{\rm hi}|\,\|R_{\rm lo}\|_t/\|y_{\rm hi}\|_t$. Together: $|\langle Y_{\rm hi}^2 r_{\rm lo}\rangle|\le\kappa\|Y_{\rm hi}\|^2$. (f) $|\langle Y_{\rm hi}^2 r_{\rm hi}\rangle|\le\eta_{\rm hi}\|Y_{\rm hi}\|^2$. Dividing by $\|Y\|^2$ gives $|\langle Y^2r\rangle|\le E\|Y\|^2$.

Step 4 (the square). Write $Yr=Y_{\rm lo}r_{\rm lo}+Y_{\rm hi}r_{\rm lo}+Yr_{\rm hi}$. By Minkowski,

$$\|Yr\|\le\|Y_{\rm lo}r_{\rm lo}\|+\|Y_{\rm hi}r_{\rm lo}\|+\|Yr_{\rm hi}\|. \tag{D7}$$

Here $\|Y_{\rm lo}r_{\rm lo}\|\le\eta_{\rm lo}\sqrt{f_{\rm lo}}\|Y\|$ and $\|Yr_{\rm hi}\|\le\eta_{\rm hi}\|Y\|$. The middle term uses the same two bounds as (e): $\|y_{\rm hi}R_{\rm lo}\|_t\le\eta_{\rm lo}\|y_{\rm hi}\|_t$ and $\|y_{\rm hi}R_{\rm lo}\|_t\le\sup|y_{\rm hi}|\,\|R_{\rm lo}\|_t$, so $\|Y_{\rm hi}r_{\rm lo}\|\le\kappa\sqrt{f_{\rm hi}}\|Y\|$. Hence $\|Yr\|^2\le B\|Y\|^2$; the same κ enters E and B.

Step 5 (Lemma S). Steps 2–4 give $\|Y\|^2(1-E)\le\langle Y^2/h\rangle\le\|Y\|^2(1+E+B/(1-\eta))$. The η sandwich $1-\eta\le h\le1+\eta$ gives the other two bounds; take the better of each pair.

Lemma S′. Without the gap, (b) and (c) are bounded instead of vanishing: $|\langle Y_{\rm lo}^2 r_{\rm hi}\rangle|\le\eta_{\rm hi}\|Y_{\rm lo}\|^2$ and $2|\langle Y_{\rm lo}Y_{\rm hi}r_{\rm lo}\rangle|\le2\eta_{\rm lo}\|Y_{\rm lo}\|\|Y_{\rm hi}\|$. These add $\eta_{\rm hi}f_{\rm lo}+2\eta_{\rm lo}\sqrt{f_{\rm lo}f_{\rm hi}}$ to E. Step 1 needs only $M<m$, and Step 4 uses no gap, so B is unchanged.

Where Lemma S is used. Lemma S is a statement about period averages. In the windows of this paper the v-average is Gaussian, i.e. the period average weighted by θ_τ (Lemma 19 of [1]). Lemma R first converts every Gaussian average into a period average, with the explicit ν factors of Lemmas 21 and 22 of [1]: $J_{ww}=J_{\rm per}(1+\epsilon_1)$, $|\epsilon_1|\le\nu_0(1+2\bar Q)/(1-z)$, and the two-sided bound of W by $W_{\rm per}$ with $(1\pm\nu_0)$ and $\pm\frac12 b\tau\bar Q\nu_2$. Only then is a sandwich applied, to three period averages, each of the form of Lemma S with $\psi=E=e^{-w^2/2s^2}$:

- $J_{{\rm per},R}=\int\Gamma\langle h_w^2/h\rangle$: $y=R$, $\phi=-(w/s^2)E$, $dm=\Gamma\,dw$;
- $W_{\rm per}=x\int\Gamma\langle h_t^2/h\rangle$: $y=R'$, $\phi=E$, $dm=x\,\Gamma\,dw$;
- the F term of (R4): $y=R'$, $\phi=E$, $dm=x\,\Gamma(1-\cos\frac{w}{\sqrt2})\,dw$, a positive measure.

Lemma S replaces (R1) in each of them, and nothing else changes. The ν factors are those of certificate (ii), taken at $\tau=20$. Lemma H (App. A) is not used in this chain.

3. Lemma A4*

Kernels. With $e_n=e^{-n^2\tau/2}$, $\theta_\tau=\sum_n e_{|n|}e^{int}$ and $\Theta=\sum_n e_{|n|}e^{n\beta}e^{int}$ (Lemma 19 of [1]). So $\widehat{(\theta_\tau-\Theta)}_n = e_{|n|}(1-e^{n\beta})$, and for any (complex) g,

$$\begin{aligned}\langle(\theta_\tau-\Theta)g\rangle_t = \sum_{n\ge1} e_n\big[&(1-e^{n\beta})\hat g_{-n}\\ &+(1-e^{-n\beta})\hat g_n\big].\end{aligned} \tag{D8}$$

With $e^{\pm n\beta} = \cosh n\beta \pm \sinh n\beta$ this is

$$\sum_{n\ge1} e_n\big[(\hat g_n+\hat g_{-n})(1-\cosh n\beta) + (\hat g_n-\hat g_{-n})\sinh n\beta\big]. \quad \text{(D9)}$$

This identity does not need g to be real. Here $g = Fe^{i\psi}$ is complex, and $|\hat g_{\pm n}| \le \langle|g|\rangle_t = \langle F\rangle_t$ because $F \ge 0$.

First sum. $\cosh u - 1 \le \frac{u^2}{2}e^u$ gives at most $\sum_n e_n\, 2\langle F\rangle_t \frac{n^2\beta^2}{2}e^{n\beta} = \frac{\beta^2\nu_2}{2}\langle F\rangle_t$, with $\nu_k = 2\sum n^k e_n e^{n\beta}$.

Second sum. Write $\psi = w/\sqrt2 + \chi$ and $g = g^0 + g^1$, $g^0 = e^{iw/\sqrt2}F$, $g^1 = e^{iw/\sqrt2}F(e^{i\chi}-1)$. F is real, so $\hat F_{-n} = \overline{\hat F_n}$ and $\hat g^0_n - \hat g^0_{-n} = 2i\, e^{iw/\sqrt2}\,\mathrm{Im}\hat F_n$. Its real part is $-2\sin(w/\sqrt2)\,\mathrm{Im}\hat F_n(w)$.

Parity. In every state $h = 1 + e^{-w^2/2s^2}R(t)$, so h, h_t and $F = k'^2h_t^2/h$ are even in w, and so is $\mathrm{Im}\hat F_n(w)$. Γ is centred, e and β do not depend on w, so

$$\int \Gamma(w)\,\sin\frac{w}{\sqrt2}\,\mathrm{Im}\hat F_n(w)\,dw = 0 \quad \text{(D10)}$$

for every n and every allowed state; the g^0 part of the second sum vanishes after integration in w.

The rest. $|g^1| \le F|\chi| \le F\bar\chi(w)$, so $|\hat g^1_n - \hat g^1_{-n}| \le 2\bar\chi\langle F\rangle_t$, and $\sinh u \le ue^u$ gives at most $\beta\nu_1\bar\chi(w)\langle F\rangle_t$.

The pieces of $\bar\chi$. $\chi = \sigma^2(\tilde s + s_w)/\sqrt2$ with $\sigma^2|\tilde s| = \sigma^2k'|h_t|/h \le \xi = \sigma\sqrt{x}N_1/(1-\bar Q)$ and $\sigma^2|s_w| \le \sigma^2|w|/L^2 + \zeta$, $\zeta = \sigma^2\eta e^{-1/2}/(s(1-\eta))$ (from $\sup_w |w|e^{-w^2/2s^2}/s^2 = e^{-1/2}/s$). So $\bar\chi = (\xi + \zeta + \sigma^2|w|/L^2)/\sqrt2$.

Integration in w. $e\sigma^2\int\Gamma\langle F\rangle_t = eW_{\rm per}$. The weight $\langle F\rangle_t \propto e^{-w^2/s^2}$ turns Γ into a centred Gaussian of variance s_{e2}, whose mean of $|w|$ is $\sqrt{2s_{e2}/\pi}$; so the $|w|$ part gives $\sigma^2\sqrt{s_{e2}/\pi}/L^2$. The η sandwich bounds $W_{\rm per}$ by $W_{\rm per,up}$. This is a_4^*. Every piece is $W_{\rm per}$ times $\beta^2\nu_2$ or $\beta\nu_1\sigma$: second order in β, or first order in β and in σ.

4. Lemma Z3′

Setting. (P2) and $K \ge x$ give $q0_n \le q0_1 d_n$, $d_n = e^{-(n^2-1)x/2}$; (P3) gives $\sum q0_n \le \bar Q$. Put $y = q0_1$, $v_n = q0_n/y \in [0,d_n]$ $(n \ge 2)$, $s = \sum v_n^2$, $t = \sum v_n$. Then $S_0 = y^2(1+s)$, $P = \mu S_0 = \mu y^2(1+s)$ and $y(1+t) \le \bar Q$, so

$$P \le \bar Q^2\mu\,\frac{1+s}{(1+t)^2}. \quad \text{(D11)}$$

(i) Mean identity. $\mu(1+s) = x(1+\sum n^2v_n^2)$, i.e. $\sum_{n\ge2} v_n^2(n^2x-\mu) = \mu - x$. On $H = \{n^2x > \mu\}$, $v_n^2(n^2x-\mu) \le v_nc_n$ with $c_n = d_n(n^2x-\mu) = e^{(x-\mu)/2}ze^{-z/2} \le \frac{2}{e}e^{(x-\mu)/2}$, $z = n^2x-\mu$.

(ii) $s \le a\,t$ with $a = c_s(x)e^{-\mu/2}$, $c_s(x) = e^{x/2}(1+\sqrt{e+2/e})$. On H, $v_n^2 \le d_nv_n \le e^{(x-\mu)/2}v_n$. On $\{0 \le \mu - n^2x \le 1\}$, $v_n \le d_n \le e^{(x-\mu+1)/2}$. On $\{\mu - n^2x > 1\}$, $\sum v_n^2 \le \sum v_n^2(\mu - n^2x) \le \sum_H v_nc_n \le \frac{2}{e}e^{(x-\mu)/2}t$ by (i).

(iii) By (i), $\mu - x \le \sum_H v_nc_n \le \frac{2}{e}e^{(x-\mu)/2}t$, so $t \ge t_0(x) = \frac{e}{2}(\mu-x)e^{(\mu-x)/2}$.

(iv) Monotonicity. $\frac{d}{dt}\frac{1+at}{(1+t)^2} = \frac{a-2-at}{(1+t)^3}$. If $a \le 2$ this is negative for every $t \ge 0$; so $\frac{1+s}{(1+t)^2} \le \frac{1+at}{(1+t)^2} \le \frac{1+at_0}{(1+t_0)^2}$.

Removing x. For $x \le 0.003 < \mu$, $t_0(x)$ decreases in x (it is increasing in $\mu - x$), so $t_0(x) \ge t_0(0.003)$. $c_s(x)$ increases in x, so $a \le c_s(0.003)e^{-\mu/2}$; and the bound of (iv) increases in a. This gives $\Pi^*(\mu)$, with $c_s = c_s(0.003) = 3.3896$. Finally $s \le t$ (as $v_n \le 1$), so $(1+s)/(1+t)^2 \le 1$ and $P \le \mu\bar Q^2$.

Range. (iv) needs $a \le 2$, i.e. $\mu \ge 2\ln(c_s/2) = 1.055$. The certificates use $\Pi^*(\mu)$ only for $\mu \ge 2$ and the bound $\mu\bar Q^2$ for $\mu < 2$ (`cc_cells.py`, `ce_link.py`). For $\mu \in (0.003, 0.226)$ the formula Π^* is not a valid bound by this argument, because $t_0 < (a-2)/a$ there; the statement in Sec. V is restricted accordingly. The envelope values of Sec. V are attained at $\mu > 5$.

5. Lemma BE

Where the cross term enters. In the exact identity of Lemma 23 of [1], the cross term is $\sigma^2\mathrm{Re}A[h\,2\alpha\psi'e^{i\psi}]$ with $2\alpha\psi' = (1-b)\sigma^2G$. After the θ_τ steps (remainders a_5, a_6) it is $e(1-b)\,\sigma^2\int\Gamma\langle h\,\sigma^2G\cos\psi\rangle$, divided by $Z \ge 1-z$. The factor σ^2 in front is the one that produces $-\sigma^2\gamma_P\Phi$ in Lemma 23 of [1].

(BE-1) applied. With (BE-1) and $e(1-b) \le e$, this term is at least

$$-\,e(1-b)\sigma^4\int\Gamma\langle F\rangle \;-\; e\sigma^2\Big[\frac{M_G}{4c} + 2c\,\Phi\Big]. \quad \text{(D12)}$$

The first part is the $-\sigma^2W$ of Lemma 23 of [1]; the bracket replaces $\gamma_P\Phi$.

The choice of c. With $c = (b+\sigma^2J_{ww})/(16\sigma^2)$:

$$e\sigma^2\cdot 2c\,\Phi = \frac{e(b+\sigma^2J_{ww})\Phi}{8} \le \frac{b\Psi}{2} + \frac{\sigma^2J_{ww}}{4}, \quad \text{(D13)}$$

by $\Phi \le 4\Psi$ (Lemma 23 of [1]) and $\Phi \le 2$; and

$$e\sigma^2\frac{M_G}{4c} = \frac{4e\sigma^4M_G}{b+\sigma^2J_{ww}}. \quad \text{(D14)}$$

The M_G part. By (BE-2) with $x^2S_4 \le \Lambda xS_2 + T_\Lambda$,

$$M_G \le \frac{3g_s}{2}\Big[xS_2K_1 + \frac{T_\Lambda}{1-\bar Q}\Big]. \quad \text{(D15)}$$

Ripple piece. Use $xS_2g_s \le 2(1+\bar Q)W_{\rm per}$ and $b+\sigma^2J_{ww} \ge b$:

$$\frac{4e\sigma^4}{b}\cdot\frac{3}{2}\cdot 2(1+\bar Q)K_1W_{\rm per} = \frac{12(1+\bar Q)e\sigma^4K_1}{b}W_{\rm per}. \quad \text{(D16)}$$

With $1-e \geq y/(2+y)$, $y = (1-b)^2\sigma^2/(2b)$, i.e. $\sigma^2/(by) = 2/(1-b)^2$,

$$\frac{12(1+\bar{Q})e\sigma^4 K_1}{b(1-e)} \leq 12(1+\bar{Q})K_1\sigma^2\frac{2}{(1-b)^2} \times e^{-y/2}(2+y) \leq \frac{48(1+\bar{Q})K_1\sigma^2}{(1-b)^2} = \rho_W, \quad \text{(D17)}$$

since $e^{-y/2}(2+y) \leq 2$. With $\Lambda = 10^{-3}/\sigma^2$, $\sigma^2 K_1 = 10^{-3}/(1-\bar{Q}) + \sigma^2(U_1^2+\frac{1}{2})/(1-\bar{Q})^3$. The second term is below 10^{-40} for $\sigma \leq 10^{-24}$ and $U_1 \leq 29.4$, so $\rho_W = 0.08 + O(10^{-40}) \leq 0.0801$.

Tail piece. Use $b + \sigma^2 J_{ww} \geq \sigma^2 J_{ww}$:

$$\frac{4e\sigma^4}{\sigma^2 J_{ww}} \cdot \frac{3g_s}{2} \cdot \frac{T_\Lambda}{1-\bar{Q}} \leq \frac{6\sigma^2 g_s T_\Lambda}{(1-\bar{Q})J_{ww}}. \quad \text{(D18)}$$

The factor σ^2 stays: it is the σ^2 of the cross term times $\sigma^4/(\sigma^2 J_{ww})$. We compare it with $\sigma^2 J_{ww}$, where $J_{ww} \geq 0.6/L^2 \geq 0.6\sigma$: the ratio is $8g_s T_\Lambda/J_{ww}^2 \leq 22.3\,T_\Lambda/\sigma^2$. With $T_\Lambda \leq 4e^{-\Lambda}(\Lambda^2 + \Lambda^{3/2}x^{-1/2}/1.25)$ and $x^{-1/2} \leq 1/(7\times 10^5\sigma)$ the ratio is below $10^{-4\times 10^{44}}$.

Collecting. Lemma 23 of [1], with this replacement and $\Psi \geq \min(\Phi, \frac{1}{2})$, gives

$$2N_\phi \geq (1-e)W\Big(1 - \frac{\sigma^2}{(1-e)(1-\nu_0)} - \rho_W\frac{1+z}{(1-z)(1-\nu_0)}\Big) + b\Psi\Big(\frac{1}{1+z} - \frac{1}{2(1-z)}\Big) - \frac{\sigma^2 J_{ww}}{4(1-z)} - \text{rem}. \quad \text{(D19)}$$

Here $W_{\text{per}} \leq ((1+z)W + \frac{1}{2}b\tau\bar{Q}\nu_2)/(1-\nu_0)$ was used, and $\sigma^2/(1-e) \leq \sigma^2 + 4b/(1-b)^2 \leq 4.01\times 10^4\sigma^2$. With $2A_w \geq \sigma^2 J_{ww}$, $(a) + (a') = (2A_w + N_\phi)/(\sigma\,\text{den})$, and $\nu_0, z \leq 10^{-10^7}$, this is the inequality of (BE-3).

Remainders. rem collects $a_1, a_2, a_3, a_5, \ldots, a_{10}$, a_4^*, $\frac{1}{2}b\bar{Q}(\nu_0 + \tau\nu_2)\Phi$, the ν part of the W sandwich and the tail piece. Each is a product of a power of $1/\sigma$ (at most σ^{-2} against $0.6\sigma^3$), a power of $\tau \leq$ any, and $\nu_k \leq 2.0000001e^{-0.499999\tau}$ with $\tau \geq 5\times 10^7$. $\tau^j e^{-\tau/2}$ decreases there, so the worst end is $\tau = 5\times 10^7$, $\sigma = 10^{-3000}$, and each ratio is below $10^{-1.08\times 10^7}$.

Rounding. Every printed constant is rounded in the safe direction: κ and 1.14×10^{-5} down (value 1.1500×10^{-5}), $1-e$ and y down, Ψ down (0.22120 to 0.2211), ρ_W up.

Appendix E: Specification of the cell proofs

This appendix states what a reader needs to check that the cell proofs of Sec. VI are sound: the floating-point model, the rounding bounds, the acceptance rule, the monotonicity of every term and the treatment of underflow. The cells are evaluated in double precision, not in interval arithmetic; soundness comes from an acceptance margin that is larger than a derived bound of the rounding error.

Floating-point model. IEEE 754 binary64 with round to nearest, unit roundoff $u = 2^{-53}$. We assume that log, exp, $\log 1p$, expm1 and $\sqrt{\ }$ return a value within two ulp of the exact value at the floating argument (relative error $\leq 4u$). Cell ends and ℓ are floating numbers and are exact; a cell is the set of real points between its ends, and a bisection reuses the same floating midpoint on both sides, so the cells of a column cover its domain exactly.

Certificate (i): the main reduced inequality. The test compares two logarithms, ℓ_{pos} and ℓ_{neg}, obtained from the cell ends by a chain of at most 25 operations. Each operation has error at most u (or $2u$ for an elementary function) times the size of its operands. The operands are the cell ends (each used at most twice), ℓ, constants below 100, and $\log G_{\text{cell}}$, which enters only inside a log-add-exp and is either dominant or dropped exactly. Hence, with $B = 2(|\log r_a| + |\log r_b| + |\log b_a| + |\log b_b| + |\log \mu_a| + |\log \mu_b|) + \ell_b + 100$,

$$|\hat{\ell}_{\text{pos}} - \ell_{\text{pos}}| + |\hat{\ell}_{\text{neg}} - \ell_{\text{neg}}| \leq \delta = 60u\,(|\hat{\ell}_{\text{pos}}| + |\hat{\ell}_{\text{neg}}| + B). \quad \text{(E1)}$$

A cell is accepted if $\hat{\ell}_{\text{pos}} \geq \hat{\ell}_{\text{neg}} + \max(10^{-12}, S\delta)$ with the safety factor $S = 10$ (`cl_cells.py`).

Certificates (ii) and (iii): the full Lemma R bound.* Every term is e^{l_t}, where l_t is a sum of at most 16 operands: cell ends times coefficients $|c| \leq 2$, powers $p\ell$ with $p \leq 3$, constants, $\log G$ and logarithms of monotone factors. By Higham's bound, $|\hat{l}_t - l_t| \leq \gamma_{k+4}\sum|x_i|$, and $\sum|x_i| \leq |l_t| + 2B$ with $B = 2(\max|\log\mu| + \max|\log r| + \max|\log b|) + 3\ell_b + 600$. The factor $e^{-y/2}$ adds $4u\,y_{\max}B$ (if $y_{\max} \geq 1500$ it underflows to zero and the positive term is dropped), and the later steps (netting at equal powers of σ, the shift by $p\log\sigma$, normalisation) add $4u(|l_t| + B + 3\ell_b)$. So

$$|\hat{l}_t - l_t| \leq \eta_t = 25u\,(|l_t| + 3B + y_c B). \quad \text{(E2)}$$

The remaining operations (sums of at most 60 terms, netting, the closed-form minimum in a', square roots) act on quantities normalised by the largest part. Their error is at most $1000u$ times the *unnetted* magnitude, i.e. the sum of the absolute values of all terms. This is the point that a rule of the form "value/magnitude $> 10^{-10}$" after netting does not cover: when equal powers of σ nearly cancel, the netted magnitude can be smaller than the unnetted one by the ratio R ($R \leq 7\times 10^4$ was observed).

The rule. Before any other step, every positive term is lowered to $l_t - S\eta_t - 1000Su$ and every negative term is raised to $l_t + S\eta_t + 1000Su$, with $S = 10$ (`cl_link.py`). These perturbed terms bound the exact ones in the safe direction, and the margin $1000Su$ covers the rounding of all later steps relative to the unnetted magnitude. A cell is accepted if the resulting normalised minimum exceeds 10^{-10}. Acceptance therefore implies that the exact lower bound of the cell is positive.

A check of the bounds. Recomputing $\ell_{\text{pos}} - \ell_{\text{neg}}$ in long double (64-bit mantissa) on random cells gives differences

of 6.8×10^{-13} at $\ell = 961$ and 2.7×10^{-12} at $\ell = 6898$, below δ (`ck_rounding.py`) [numerical].

Recomputation. All certificates were recomputed with these rules on exactly the stored columns. All 92 columns of (ii), all 17 windows of (iii) and all 987 columns of (i) close. The cell counts are those of the earlier rule in all but 127 columns, where at most 176 more boxes are needed. In (i) the smallest ratio of the accepted margin to $S\delta$ is 1.00. In (ii), at $\ell = 55.26$ the column still closes with $S = 10^7$.

The τ cells and Lemma BE. The factor of App. A is built from at most 80 exponentials, one sum, a few field operations and one square root, so its relative rounding error is at most $20u$. Its margin 8.6×10^{-6} above 0.99977 is more than 3×10^9 times this bound, and a 50-digit decimal evaluation agrees to within 3.5×10^{-16}. The constants of Lemma BE are computed in 60-digit decimal; there is no double-precision acceptance rule there (`cl_tau_be.py`).

Monotonicity. On a cell, every term is bounded by its value at one corner. The direction is fixed by the following table, where "+" means that the term is evaluated at the end that makes it smallest if it is positive and largest if it is negative.

factor	monotone in, and why
σ^p, $p \geq 0$	decreasing in ℓ; positive parts at σ_{lo}, negative at σ_{hi}, after netting at equal σ
$\gamma = T_D/\sqrt{\ell}$	decreasing in ℓ; taken at ℓ_a
$y = (1-b)^2\sigma^2/(2b)$	decreasing in b, increasing in σ
$e^{-y/2}$, $1/(1+w_0)$	decreasing in y; positive terms at $y_{\max}$
$(1-b)^2$	decreasing in b
r, $\sqrt{r/(2+r)}$, $1/(2+r)$	monotone in r (powers and ratios of linear functions)
$\mu^{\pm 1}$, G/μ	monotone in μ
$G_{\mathrm{cell}} = \Pi^*(\mu) g_s(r)$	Π^* decreasing for $\mu \geq 2$ (left end), $\mu\bar{Q}^2$ for $\mu < 2$ (right end); g_s increasing in r
b, $\sqrt{b}$, $1/b$	monotone in b
$\tau^j \nu_k$	decreasing in $\tau \geq 20$ (checked on $[20, 400]$); taken at $\tau_{\min}$
U_1, U_2, γ_P, c_χ	increasing in ℓ; taken at ℓ_b

The main reduced function is concave in G, and the Lemma R* bound is affine in G for each A, so $G \in \{0, G_{\mathrm{cell}}\}$ suffices (Sec. V).

Underflow and clipping. All terms are carried as logarithms and normalised by the largest part before exponentiation. Exponents are clipped at 700 for positive parts (clipping lowers them) and at 709 for negative parts, and a negative part is never rounded down to zero (a floor of 10^{-300} is added). Underflow of a positive part only drops a non-negative quantity. Underflow of $e^{-y/2}$ drops the positive term that carries it.

[1] T. Namba, *How fine can a hidden ripple be? A two-sided logarithmic law for the resolution gap of the arrow of time*, v1.5, Zenodo (2026), doi:10.5281/zenodo.23856796.